

BUSINESS ANALYTICS

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1 Introduction

1.1 Aims and Objectives

The aim of this study is to analyze the flight booking dataset gotten from “Ease My Trip” website to gain insights into the factors influencing flight ticket prices and to develop a predictive model using linear regression to accurately estimate ticket prices based on the class of the flight. Logistics Regression model will also be used to classify the number of stops. By understanding the relationships between these variables and ticket prices, we aim to provide valuable information for both consumers and the stakeholders, ultimately improving decision-making processes and optimizing pricing strategies within the aviation industry.

The objective of the study is to:

- Explore Factors Influencing Ticket Prices: Identify and analyze the key factors that influence flight ticket prices, including airline, class, number of stops, departure/arrival times, and duration.
 - Visualize Price Variations: Visualize the variations in ticket prices based on different categorical variables such as airline, class, departure/arrival times, and number of stops to understand the patterns and trends.
 - Evaluate Impact of Time: Evaluate the impact of time-related factors such as days left before departure, departure/arrival times on ticket prices, and analyze how these variables affect pricing strategies.
 - Develop Predictive Model: Develop a predictive model using linear regression to accurately estimate flight ticket prices based on the class of the flight and evaluate its performance using appropriate metrics such as mean squared error, mean absolute error, and R-squared.
- Offer actionable insights and recommendations to stakeholders in the aviation business, including airlines and consumers, using descriptive, diagnostic, predictive, and prescriptive analysis. This information can help improve competitiveness and profitability by informing pricing plans, marketing initiatives, and customer decision-making processes (Bisht, 2024).

The study will attempt to answer the following business questions:

- Does price vary with Airlines?
- How is the price affected when tickets are bought in just 1 or 2 days before departure?
- Does ticket price change based on the departure time and arrival time?
- How does the ticket price vary between Economy and Business class?
- How does the number of stops affect ticket prices?

1.2 Description of the Flight Dataset

The dataset used for this study contains information about flight booking options from the Easemytrip website for flight travel between India's top 6 metro cities. The dataset contains 300,261 rows and 11 column and is stored in a comma separated value format. This dataset was obtained from Kaggle website (*Flight Price Prediction*, 2022).

Feature	Description	Data Type
Airline	Name of the airline company	Categorical
Flight	Flight code	Categorical
Source City	City from which the flight takes off	Categorical
Departure Time	Time periods grouped into bins representing departure time	Categorical
Stops	Number of stops between source and destination cities	Categorical
Arrival Time	Time intervals grouped into bins representing arrival time	Categorical
Destination City	City where the flight will land	Categorical
Class	Seat class (Business or Economy)	Categorical
Duration	Overall amount of time it takes to travel between cities (hours)	Continuous
Days Left	Difference between trip date and booking date	Continuous
Price	Ticket price	Continuous

TABLE 1 – DESCRIPTION OF KEY FEATURES IN THE FLIGHT DATASET.

1.3 Analytics Design

- **Data Collection and Integration:** We obtained the flight booking dataset from Kaggle which was originally gotten from Ease My Trip website using web scraping techniques. The data collected included information such as airline, flight code, source city, departure time, stops, arrival time, destination city, class, duration, days left, and price. The collected data was integrated into a structured format suitable for analysis.
- **Data Preprocessing:**
 - Data Cleaning:** We performed data cleaning to handle missing values and ensure data integrity.
 - Data Transformation:** Certain variables were transformed, such as converting flight duration from hours to minutes and converting prices from Indian rupees to US dollars.
 - Data Sub setting:** We focused our analysis on specific features relevant to our research questions (Jansen et al., 2023).
- **Data Analytics:**

Exploratory Data Analysis (EDA): We conducted EDA to gain insights into the relationships between different variables in the dataset. Visualization techniques such as bar plots, line plots, box plots, and KDE plots were used to represent data distributions and identify patterns.

- **Machine Learning:**

Linear Regression Modelling: We employed linear regression to predict flight prices based on the class of the flight which exhibits a linear relationship (Qu, 2024).

Logistic Regression Modelling: For this machine learning model, the independent variables or predictors were the price of the flights and the class of the flights.

Model Evaluation: Performance metrics such as mean squared error (MSE), mean absolute error (MAE), and R-squared were used to assess the accuracy of the predictive model.

- **Reporting and Visualization:** The findings of our analysis were presented in a comprehensive report to stakeholders. Visualizations such as bar plots, line plots, box plots, and KDE plots were included in the report to illustrate key insights and trends.

Flight Data Analytics – UML Activity Diagram

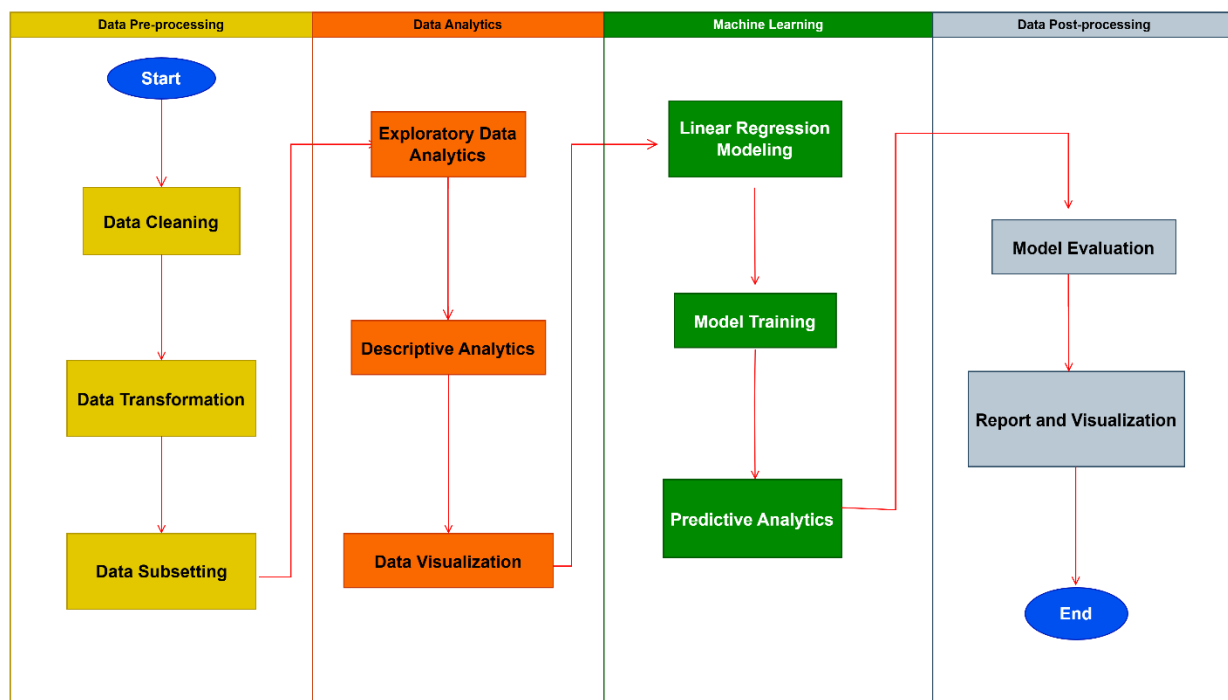


FIGURE 1 – UML ACTIVITY DIAGRAM FOR FLIGHT ANALYTICS DESIGN.

2 EXPLORATORY DATA ANALYSIS

The analysis carried out in this section was completed using both Python and SAS university edition. The code used to produce the results displayed can be seen in the Appendix section of this report.

2.1 Data Pre-processing

At the stage of data pre-processing a range of techniques were applied to the original flight dataset in preparation for the machine learning activities.

2.1.1 Data Cleaning

We began by cleaning the dataset to handle missing values and ensure data integrity. No significant data cleaning was required as the dataset was relatively clean.

2.1.2 Data Transformation

We transformed the hour variable to duration minutes which was attained by multiplying the hours by 60. We also converted the price column which was originally in Indian rupee to United States Dollars (USD), which is an international currency thereby making it easier for the stakeholders to gauge the performance of each airline and makes them more suitable for analysis. It is important to note that we used the rate of 1 Indian rupee to \$0.012 as of the 24th of April 2024.

Obs	VAR1	airline	flight	source_city	departure_time	stops	arrival_time	destination_city	class	duration_hour	days_left	Price_Indian_Rupee	duration_minutes	Price_US_dollars
1	0	SpiceJet	SG-8709	Delhi	Evening	zero	Night	Mumbai	Economy	2.17	1	5953	130.2	\$71.44
2	1	SpiceJet	SG-8157	Delhi	Early_Morning	zero	Morning	Mumbai	Economy	2.33	1	5953	139.8	\$71.44
3	2	AirAsia	I5-764	Delhi	Early_Morning	zero	Early_Morning	Mumbai	Economy	2.17	1	5956	130.2	\$71.47
4	3	Vistara	UK-995	Delhi	Morning	zero	Afternoon	Mumbai	Economy	2.25	1	5955	135.0	\$71.46
5	4	Vistara	UK-963	Delhi	Morning	zero	Morning	Mumbai	Economy	2.33	1	5955	139.8	\$71.46
6	5	Vistara	UK-945	Delhi	Morning	zero	Afternoon	Mumbai	Economy	2.33	1	5955	139.8	\$71.46
7	6	Vistara	UK-927	Delhi	Morning	zero	Morning	Mumbai	Economy	2.08	1	6060	124.8	\$72.72
8	7	Vistara	UK-951	Delhi	Afternoon	zero	Evening	Mumbai	Economy	2.17	1	6060	130.2	\$72.72
9	8	GO_FIRST	G8-334	Delhi	Early_Morning	zero	Morning	Mumbai	Economy	2.17	1	5954	130.2	\$71.45
10	9	GO_FIRST	G8-336	Delhi	Afternoon	zero	Evening	Mumbai	Economy	2.25	1	5954	135.0	\$71.45
11	10	GO_FIRST	G8-392	Delhi	Afternoon	zero	Evening	Mumbai	Economy	2.25	1	5954	135.0	\$71.45
12	11	GO_FIRST	G8-338	Delhi	Morning	zero	Afternoon	Mumbai	Economy	2.33	1	5954	139.8	\$71.45
13	12	Indigo	6E-5001	Delhi	Early_Morning	zero	Morning	Mumbai	Economy	2.17	1	5955	130.2	\$71.46
14	13	Indigo	6E-6202	Delhi	Morning	zero	Afternoon	Mumbai	Economy	2.17	1	5955	130.2	\$71.46
15	14	Indigo	6E-549	Delhi	Afternoon	zero	Evening	Mumbai	Economy	2.25	1	5955	135.0	\$71.46
16	15	Indigo	6E-6278	Delhi	Morning	zero	Morning	Mumbai	Economy	2.33	1	5955	139.8	\$71.46
17	16	Air_India	AI-887	Delhi	Early_Morning	zero	Morning	Mumbai	Economy	2.08	1	5955	124.8	\$71.46
18	17	Air_India	AI-665	Delhi	Early_Morning	zero	Morning	Mumbai	Economy	2.17	1	5955	130.2	\$71.46
19	18	AirAsia	I5-747	Delhi	Evening	one	Early_Morning	Mumbai	Economy	12.25	1	5949	735.0	\$71.39
20	19	AirAsia	I5-747	Delhi	Evening	one	Morning	Mumbai	Economy	16.33	1	5949	979.8	\$71.39

FIGURE 2 – CONVERSION OF DURATION OF FLIGHTS FROM HOURS TO MINUTES AND PRICES FROM INDIAN RUPEES TO US DOLLARS.

2.1.3 Data Sub-setting

We focused our analysis on specific features relevant to our research questions, such as airline, departure time, arrival time, source city, destination city, class, duration, days left, stops, and price.

2.2 Exploratory Data Analysis (EDA):

Through EDA, we gained insights into the relationships between different variables in the dataset. This included analyzing the distribution of flight prices across various factors such as airlines, departure time, arrival time, source city, destination city, class, and number of stops. This section aims at answering the research questions which will be answered in detail below.

2.2.1 Question 1: Does price vary with Airlines?

To answer this question, we first had to show the distribution of tickets sold by each airline and the average, lowest and highest prices of tickets sold by each airline.

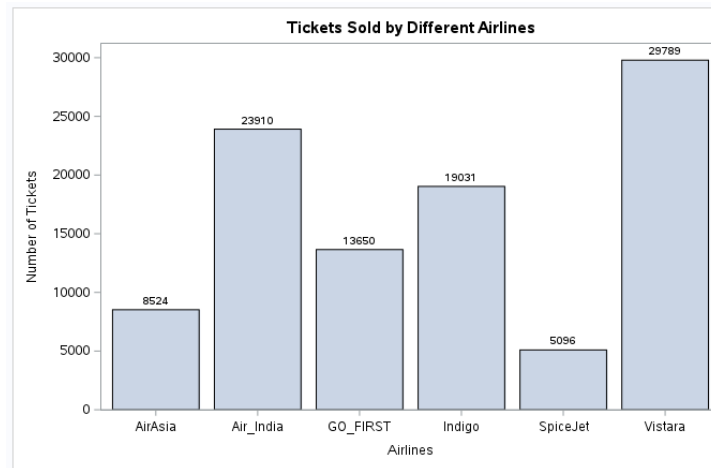


FIGURE 3: BARPLOT SHOWING THE NUMBER OF TICKETS SOLD BY THE AIRLINES

Figure 3 shows a bar chart illustrating the number of tickets sold by various airlines. The graphic reveals which airlines are most popular among people buying flights. Vistara Airlines is the top performer, selling the most tickets, followed by Air India and Indigo. By evaluating the proportion of tickets sold by each airline, we can determine their market share and competitive position in the industry. Understanding each airline's popularity can help with strategic decisions such as collaborations, marketing campaigns, and route optimization..

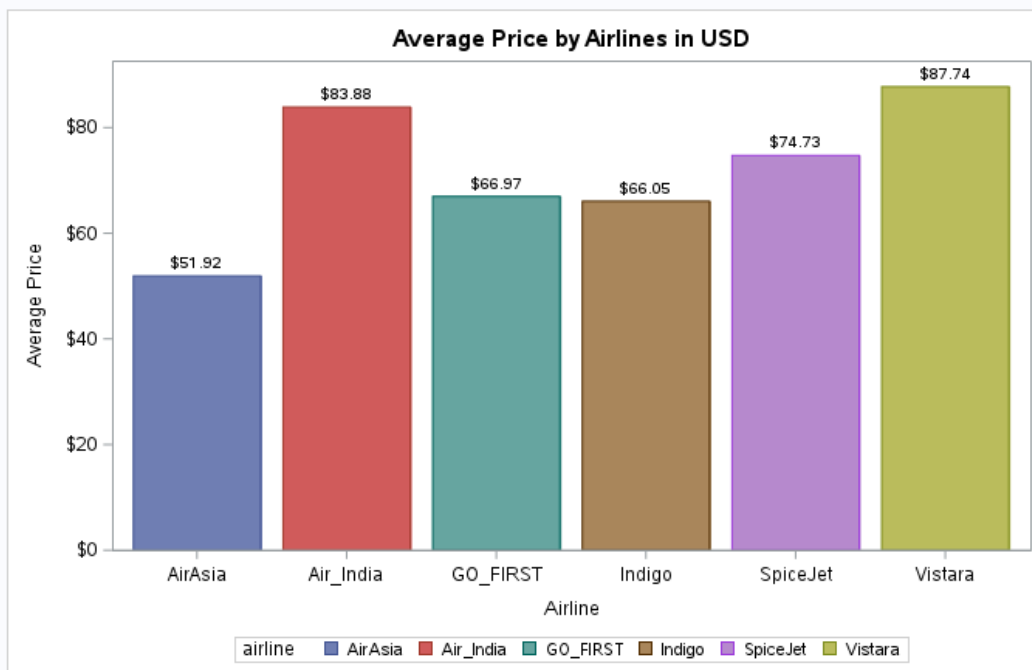


FIGURE 4 – BAR PLOT SHOWING THE AVERAGE PRICE BY AIRLINES IN USD

Figure 4 bar plot illustrates the average price of flight tickets for each airline represented in the dataset. There is noticeable variation in the average price of tickets across the different airlines. Vistara airline have higher average ticket prices followed by Air India compared to the other airlines, indicating potential differences in pricing strategies or service offerings.

This analysis provides stakeholders with insights into the pricing dynamics of various airlines operating in the market. Understanding the pricing landscape can help stakeholders make informed decisions regarding partnerships, pricing adjustments, and competitive positioning.

2.2.2 Question 2: How is the price affected when tickets are bought in just 1 or 2 days before departure.

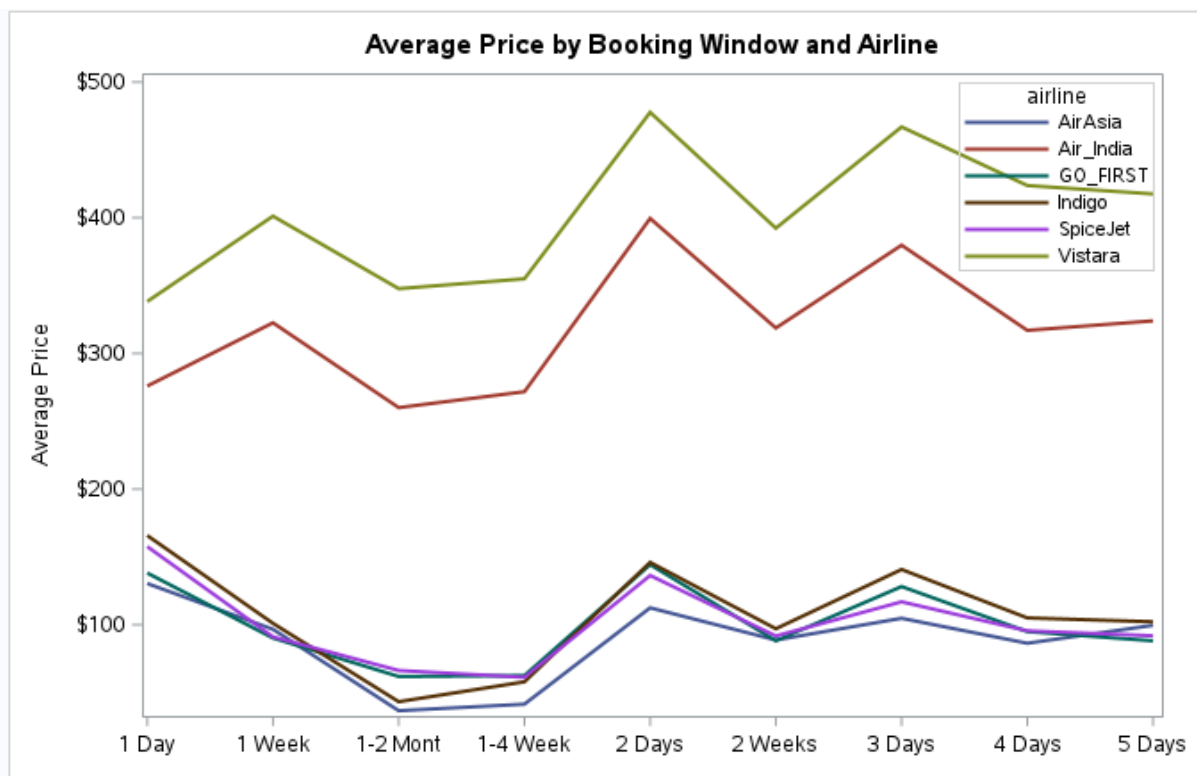


FIGURE 5 – LINE PLOT SHOWING PRICE VS. BOOKING WINDOW

The-line plot in figure 5 illustrates how ticket prices vary based on the number of days remaining before departure. Prices tend to increase significantly when tickets are purchased just one or two days before departure, irrespective of the airline. While most airlines show a similar trend of price escalation close to the departure date, Vistara and Air India stand out for offering lower prices on the last day compared to two days before departure.

The analysis highlights the impact of booking timing on flight ticket prices, with prices generally increasing significantly when tickets are purchased just one or two days before departure. By

leveraging these insights, stakeholders can develop proactive pricing strategies and promotional campaigns to optimize revenue and enhance customer satisfaction.

2.2.3 QUESTION 3: DOES TICKET PRICE CHANGE BASED ON THE DEPARTURE TIME AND ARRIVAL TIME

This question aims at finding out if the prices of the tickets change based on departure and arrival time. To do that a barplot was made to show the illustration.

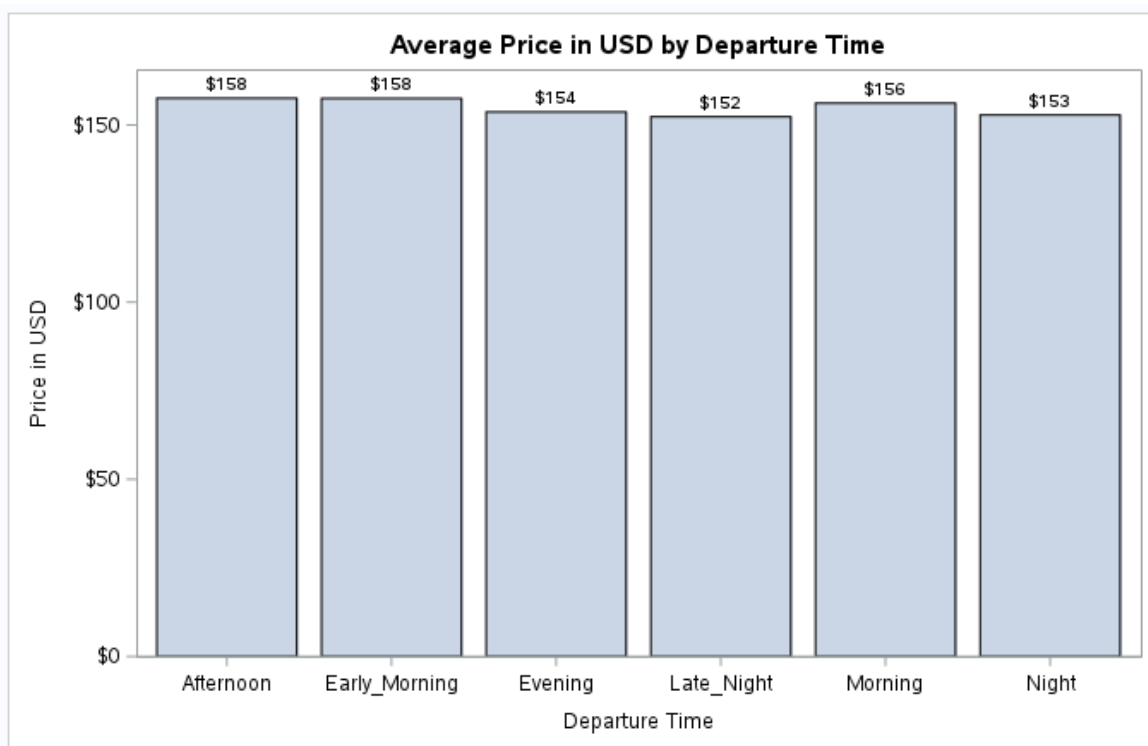


FIGURE 6: BARPLOT SHOWING THE AVERAGE FLIGHT PRICE BY DEPARTURE TIME

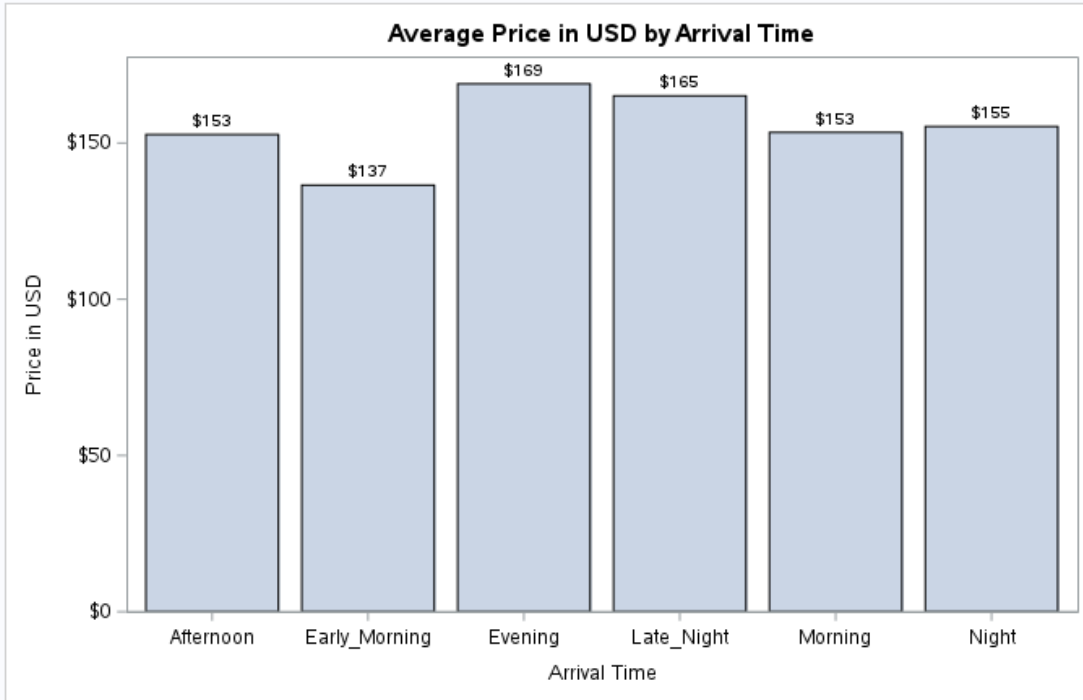


FIGURE 7: BARPLOT SHOWING THE AVERAGE FLIGHT PRICE BY ARRIVAL TIME.

From the barplot in figure 6 and 7, we can make the following observations for the departure times of the flights:

1. Cheapest flights are in the Late Night (\$152) followed by Nights (\$153).
2. Most expensive flights are afternoon (\$158) and early mornings (\$158).
3. Possible reasons for price variations include:
 - Demand: Business travel during peak hours (early mornings/afternoon) drives prices up.
 - Competition: More competition on some routes during Afternoon hours might lead to higher prices.

While for the arrival time, we can notice that flights arriving in the early mornings are cheapest at \$137, followed by late morning and afternoon (\$153). Flights that arrived at evenings were the most expensive at an average ticket price of \$169. This likely reflects business travel demand during peak hours (morning/night) and potentially higher airline operating costs for evening flights.

2.2.4 Question 4: How does the ticket price vary between Economy and Business class.

Two classes of flights were identified in the dataset used and they are both economy and business class flights. Now we need to access the variation of ticket prices based on the class of the flight.

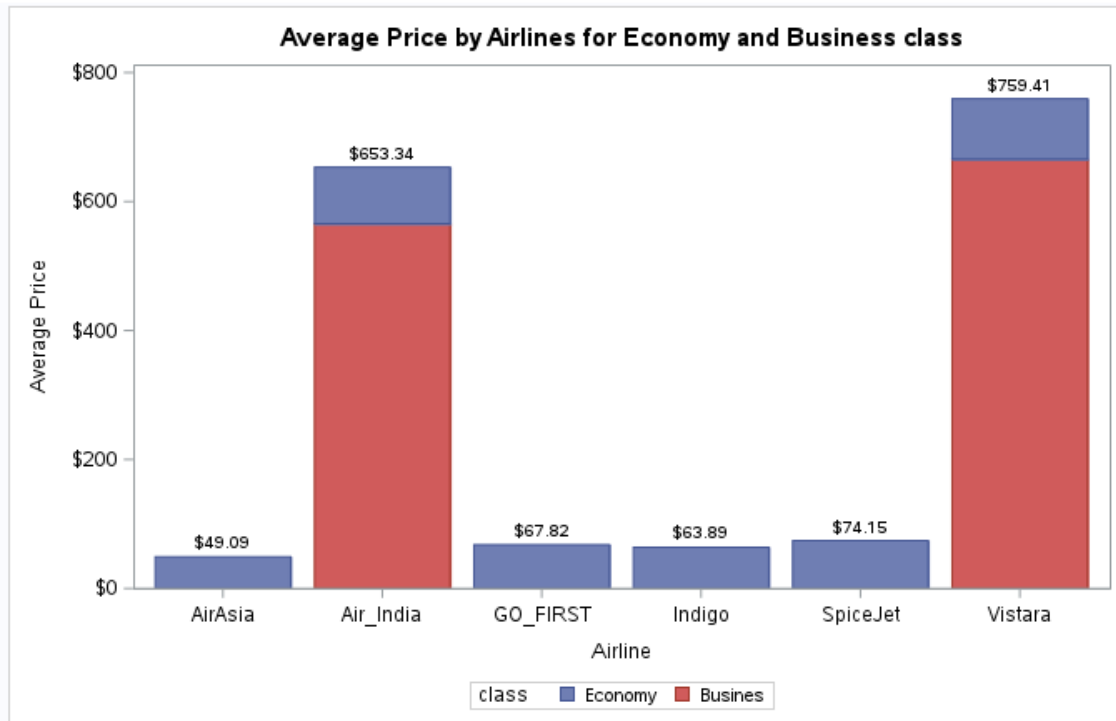


FIGURE 8: BARPLOT SHOWING THE AVERAGE FLIGHT PRICE BY AIRLINES FOR ECONOMY AND BUSINESS CLASS

From the barplot in figure 8 we can observe that only two airlines had business class. The airlines are Vistara and Air India, we can also notice that the average ticket price for the economy class were more expensive than the average ticket prices for economy class. Vistara had the most expensive of tickets prices for both economy and business classes. This could also be influenced by the fact that Vistara sold the highest number of tickets and made the highest revenue out of all the other airlines.

2.2.5 Question 5: How does the number of stops affect ticket prices.

This research questions seeks to understand the relationship between the number of stop and the prices of tickets. To achieve this, we can use a line plot to visualize the between the two variables.

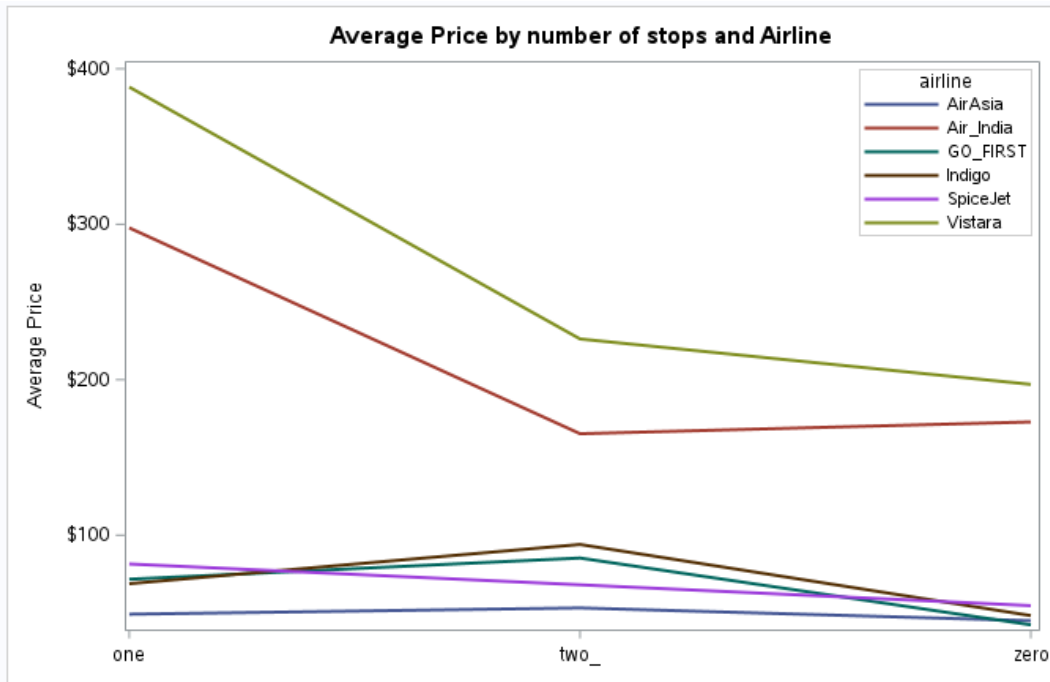


FIGURE 9: LINE PLOT SHOWING THE AVERAGE PRICE OF FLIGHT BY THE NUMBER OF STOPS FOR EACH AIRLINE

According to the graph in figure 8, As the number of stops rises, so does the average price. Accordingly, the cheapest flights are usually those with no layovers (zero stops), followed by flights with one stop, and the priciest flights are those with two stops. Even for flights with the same number of stops, there appears to be some variation in the prices offered by the carriers. Certain airlines seem more affordable than others, for instance, when it comes to zero-stop trips.

3 MACHINE LEARNING

3.1.1 Linear Regression Model

We employed linear regression modeling to predict flight prices based on class which are business and economy. The model defines a linear relationship between the two variables. This allowed us to build a predictive model that can estimate the price of a flight ticket given the class of the flight.

Linear Regression Model to Predict Price

The REG Procedure
Model: MODEL1
Dependent Variable: Price_US_dollars

Number of Observations Read	240123
Number of Observations Used	240123

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	15681321952	15681321952	1757564	<.0001
Error	240121	2142405266	8922.19034		
Corrected Total	240122	17823727218			

Root MSE	94.45735	R-Square	0.8798
Dependent Mean	250.75002	Adj R-Sq	0.8798
Coeff Var	37.66993		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	78.86186	0.23231	339.47	<.0001
class_num	1	551.81760	0.41624	1325.73	<.0001

FIGURE 10 – SHOWING LINEAR REGRESSION MODEL PRICE PREDICTION

3.1.2 Interpretation of the Result of Linear Regression Model

- The model is statistically significant ($p < 0.0001$), as the F-value is significant.
- The model explains approximately 87.98% of the variance in the dependent variable.
- For each unit increase in the class_num variable (economy to business class), there is an estimated increase of \$551.8176 in the Price_US_dollars variable, holding all other variables constant. This means that on average, the price for business class is approximately \$551.82 higher than for economy class, after controlling for the effect of other variables in the model.

3.1.3 Model Evaluation for linear Regression Model

Comparing the Predicted Prices with the Actual Prices

Obs	Price_US_dollars	predicted_price	residual
1	71.436	78.8619	-7.4259
2	71.436	78.8619	-7.4259
3	71.472	78.8619	-7.3899
4	71.460	78.8619	-7.4019
5	71.460	78.8619	-7.4019
6	71.460	78.8619	-7.4019
7	72.720	78.8619	-6.1419
8	72.720	78.8619	-6.1419
9	71.448	78.8619	-7.4139
10	71.448	78.8619	-7.4139
11	71.448	78.8619	-7.4139
12	71.460	78.8619	-7.4019
13	71.460	78.8619	-7.4019
14	71.460	78.8619	-7.4019
15	71.460	78.8619	-7.4019
16	71.460	78.8619	-7.4019
17	71.460	78.8619	-7.4019
18	71.448	78.8619	-7.4139
19	71.448	78.8619	-7.4139
20	71.448	78.8619	-7.4139
21	71.460	78.8619	-7.4019
22	73.980	78.8619	-4.8819
23	80.280	78.8619	1.4181
24	79.020	78.8619	0.1581
25	106.428	78.8619	27.5661
26	123.120	78.8619	44.2581
27	125.640	78.8619	46.7781
28	126.900	78.8619	48.0381
29	130.056	78.8619	51.1941

FIGURE 11: COMPARING THE PREDICTED PRICES AND ACTUAL PRICES.

In figure 11 above, there are instances where the model's predictions closely match the actual prices, as indicated by minimal price differences. However, there are also instances where the model deviates significantly from the actual prices, leading to notable price differences.

These differences highlight areas where the model may need further refinement or where the input features may not fully capture the variability in ticket prices.

Overall, the comparison provides insights into the model's performance and areas for improvement, aiding stakeholders in decision-making related to pricing strategies and model enhancement.

3.1.4 Logistic Regression Model

Logistics Regression Machine learning was used to predict the number of stops for the flight, it involves predicting if the flight had no stops or had just one or multiple stops.

Proc Logistics was used to achieve this with 2 response levels, which are 1 and 0.

Model Information	
Data Set	WORK.TRAIN
Response Variable	stop_num
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	240123
Number of Observations Used	240123

Response Profile		
Ordered Value	stop_num	Total Frequency
1	0	28809
2	1	211314

Probability modeled is stop_num=1.

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	176193.31	123013.17
SC	176203.69	123044.34
-2 Log L	176191.31	123007.17

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	53184.1348	2	<.0001

FIGURE 15: PROC LOGISTIC REGRESSION MODELLING

3.1.5 Interpretation of the Logistic Regression Model:

- Statistical significance is observed for the model ($p < .0001$).
- Price_US_dollars and class_num are significant predictors ($p < .0001$).
- Price_US_dollars has an odds ratio of 1.037, which means that the probability of stop_num = 1 increase by a factor of 1.037 for every unit rise in Price_US_dollars.
- Class_num's odds ratio is less than 0.001, suggesting a substantial inverse association with stop_num.
- With a Somers' D of 0.734 and a concordance of 86.7%, the model has strong predictive potential. The statistics demonstrate a robust correlation between the anticipated likelihood and the observed outcomes.

3.1.6 Model Evaluation for Logistic Regression

Residuals vs. Predicted Values				
Obs	stops_num	l_stop_num	P_0	P_1
1	0	1	0.10088	0.89912
2	1	1	0.10108	0.89892
3	1	1	0.10108	0.89892
4	1	1	0.00745	0.99255
5	1	1	0.00724	0.99276
6	1	1	0.00629	0.99371
7	1	1	0.00517	0.99483
8	1	1	0.00456	0.99544
9	1	1	0.00456	0.99544
10	1	1	0.00264	0.99736
11	1	1	0.00217	0.99783
12	1	1	0.00158	0.99842
13	1	1	0.00039	0.99961
14	1	1	0.00034	0.99966
15	1	1	0.00028	0.99972
16	1	1	0.00000	1.00000
17	0	1	0.10084	0.89916
18	0	1	0.10084	0.89916
19	0	1	0.10084	0.89916
20	0	1	0.10084	0.89916
21	0	1	0.10084	0.89916
22	1	1	0.10108	0.89892
23	1	1	0.10108	0.89892
24	1	1	0.10084	0.89916
25	1	1	0.02016	0.97984

FIGURE 16: COMPARING THE PREDICTED NUMBER OF STOPS AND ACTUAL STOPS.

From the figure above we can notice that the logistic model was successfully able to predict the number of stops in most instances. The Model couldn't make the right prediction all the time, but we can see that we have more right predictions than wrong predictions.

4.1 CRITICAL COMPARISON OF PYTHON AND SAS

4.2 Introduction

The analysis done in this project has been carried out using two different analytics tools: SAS University Edition and Python. This section provides a critical comparison of the two analytics

tools regarding the analysis done in Section 3 of this project. The comparison will cover a range of criteria including tool's efficiency, user-friendliness, scalability for large datasets, ease of use, and data visualization capabilities.

4.3 SAS University Edition Overview

The acronym SAS stands for Statistical Analysis Software. It was established in 1960 by the SAS Institute. SAS has been used for data management, corporate intelligence, predictive analysis, descriptive and prescriptive analysis, and other applications since January 1, 1960.

Since then, the software has included numerous new statistical methods and components (SAS - Overview, 2024). SAS University Edition is a free version of SAS software intended for educational use. It provides access to essential SAS products and tools like as SAS Studio, SAS/STAT, and SAS/IML via a virtual machine that can be installed on personal computers. Users can study data analysis, statistical modelling, and predictive analytics with substantial learning resources and community assistance.

4.4 Python Overview

Python is a high-level scripting language that can be interpreted, interactive, and object-oriented. Python is supposed to be very readable. It typically uses English terms instead of punctuation, and it has fewer syntactical structures than other languages.

Python is an open-source, cross-platform programming language. It is accessible under the Python Software Foundation License (which is compatible with the GNU General Public License) on all major operating system platforms, including Linux, Windows, and Mac OS.

The Python Enhancement Proposal (PEP) procedure was created to make it easier to add new features while maintaining readability. This approach enables anyone to propose a PEP for a new feature, library, or other enhancement.

Python's design philosophy stresses simplicity, readability, and clarity (Python Tutorial, 2024).

4.5 Evaluation of SAS University Edition and Python

An in-depth evaluation of SAS University Edition and Python is presented in this section. The evaluation is done subject to a range of criteria under tool's efficiency, user-friendliness, scalability for large datasets, ease of use, and data visualization capabilities.

Tool's Efficiency

- SAS University Edition: SAS is known for its efficiency in handling large datasets and complex analytical tasks. It utilizes optimized algorithms and parallel processing capabilities to deliver fast performance.
- Python: Python's efficiency depends on the libraries and packages used for data analysis. Libraries like NumPy and pandas offer efficient data manipulation, while libraries like scikit-learn provide efficient machine learning algorithms.

User-friendliness

- SAS University Edition: SAS Studio, the interface used in SAS University Edition, provides a user-friendly environment with point-and-click features, drag-and-drop functionality, and intuitive menus. However, the learning curve for the SAS programming language can be steep for beginners.
- Python: Python is generally considered user-friendly due to its simple syntax and readability. Jupyter Notebooks, commonly used for data analysis in Python, offer an interactive and user-friendly interface where users can mix code with explanatory text and visualizations.

Scalability for large datasets

- SAS University Edition: SAS is designed to handle large datasets efficiently, with features like in-memory processing and distributed computing options. SAS Viya, the cloud-native version of SAS, further enhances scalability by leveraging cloud resources.
- Python: Python's scalability for large datasets depends on the libraries and tools used. While libraries like pandas can handle moderately large datasets, Python may struggle with extremely large datasets compared to SAS.

Ease of Use

- SAS University Edition: SAS Studio offers a comprehensive environment with built-in tools for data manipulation, statistical analysis, and reporting. However, the proprietary nature of SAS and the complexity of its programming language may pose challenges for novice users.
- Python: Python's simplicity and versatility make it easy to learn and use for a wide range of tasks, including data analysis. The availability of extensive documentation, tutorials, and online resources further enhances its ease of use.

Data Visualization Capabilities

- SAS University Edition: SAS provides robust data visualization capabilities through tools like SAS Visual Analytics and SAS Visual Statistics. Users can create interactive and customizable visualizations to explore and communicate insights.
- Python: Python offers a variety of libraries for data visualization, including Matplotlib, Seaborn, and Plotly. These libraries provide a wide range of plotting options and customization features, allowing users to create publication-quality visualizations.

To quantify these observations, a score out of 10 has been allocated to each criterion for each tool in Table 2 below:

Criteria	SAS University Edition	Python
Tool's Efficiency	9	7
User-Friendliness	8	9
Scalability for Large Datasets	9	7
Ease of Use	9	9
Data Visualization Capabilities	9	8
Availability of Libraries	8	10
Community Support	8	9
Learning Resources	10	9
Integration with Other Tools	8	9
Flexibility and Customization	9	9
Deployment Options	8	9
Total	97	95

TABLE 2 – SAS UNIVERSITY EDITION AND PYTHON EVALUATION CRITERIA SCORING

Based on the total scores, SAS University Edition still has a slight edge over python programming language. However, the choice between the two depends on specific project requirements, user expertise, and organizational preferences.

4 CONCLUSION

4.1 Analytics Insights

Airline Preferences: The analysis reveals insights into the popularity of airlines among travelers booking flights. Vistara emerges as the top performer, followed by Air India and Indigo. Understanding airline preferences can inform strategic decisions related to partnerships, marketing campaigns, and route optimization.

Price Dynamics: Ticket prices vary significantly based on factors such as airline, class, departure/arrival times, source/destination cities, and number of stops. For instance, business class tickets exhibit notably higher average prices compared to economy class, indicating premium pricing for enhanced services. Additionally, prices tend to increase significantly when tickets are purchased just one or two days before departure, highlighting the impact of booking timing on ticket prices.

Geographical Influence: The analysis demonstrates variations in ticket prices based on source and destination cities. Factors such as distance, popularity of routes, and demand-supply dynamics influence these price differentials. Understanding the pricing dynamics based on geographical factors can help airlines optimize pricing strategies and revenue management practices.

Class Preferences: While economy class accounts for most ticket sales, business class represents a smaller yet significant portion of the market. Understanding the price differentials between economy and business class is crucial for revenue management and pricing strategies, allowing stakeholders to tailor pricing structures to maximize revenue potential across different passenger segments.

Impact of Stops and Duration: Non-stop flights tend to have higher average prices compared to flights with layovers or multiple stops. This trend may be attributed to the convenience and time-saving aspect of non-stop flights, which passengers are often willing to pay a premium for. Understanding the impact of stops and duration on ticket prices can inform route planning and pricing strategies.

Overall, the analytics insights provide valuable information for stakeholders in the aviation industry, including airlines, travel agencies, and consumers. By leveraging these insights, stakeholders can make informed decisions related to pricing strategies, route planning, marketing efforts, and customer service enhancements to optimize competitiveness and profitability in the market.

4.2 Limitations and Future Work

The present analysis has limitations due to the dataset's limited scope and representativeness, as well as the simplicity of the used model, which ignores external effects. To address these constraints, future research could focus on enriching the dataset with additional features such as passenger preferences and flight ratings, exploring more sophisticated modelling techniques like ensemble methods or neural networks, conducting a temporal analysis to understand pricing fluctuations over time, segmenting the dataset to uncover unique pricing patterns within different traveller demographics, and integrating external data sources such as economic indicators and social media sentiment to provide a more holistic understanding of the factors influencing ticket price

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6 Appendices

```
/* Create a Project Library */  
  
libname project '/home/u63777365/Business Analytics Assessment';  
run;  
  
/* Import the flight CSV file */  
proc import datafile='/home/u63777365/Business Analytics Assessment/Clean_Dataset.csv'  
    out=project.flight  
    dbms=csv  
    replace;  
    getnames=yes;  
run;  
  
/* Print the content of the data */  
proc contents data=project.flight;  
run;  
  
/* print the first 20 Observations of the data */  
proc print data=project.flight (obs=20) noobs ;  
run;
```

APPENDIX 1: READING THE FLIGHT CSV FILE.

```
/* Data Preparation */  
  
/*Dropping Rows with missing Values for all Variables*/  
data project.flight;  
    set project.flight;  
    if not cmiss(of _all_);  
run;  
  
/* Summary statistics for numeric variables */  
proc means data=project.flight;  
    var duration days_left price;  
run;  
  
/* Convert duration to minutes */  
data project.flight;  
/* Rename duration to duration_hour to represent hour */  
    set project.flight(rename=(duration=duration_hour));  
    duration_minutes =duration_hour*60;  
run;  
  
/* Print 20 obs of the data */  
proc print data=project.flight (obs=20);  
run;
```

APPENDIX 2: PERFORMING DATA TRANSFORMATION

```

/* Convert price from india rupee to US dollars */
data project.flight;
/* convert flight ticket price to indian Rupee */
set project.flight(rename=(price=Price_Indian_Rupee));
/* 1 Rupee = 0.012 USD as of 29/04/2024 */
Price_US_dollars = Price_Indian_Rupee*0.012;
run;

/* Summary statistics for numeric variables */
proc means data=project.flight;
var duration_hour days_left Price_US_dollar;
run;

/* Print 20 obs of the data */
proc print data=project.flight (obs=20);
run;

proc print data=project.flight (obs=20);
format Price_US_dollars dollar10.2;
run;

/* Calculate frequency of airline */
proc freq data=project.flight noprint;
tables airline / out=airline_freq (rename=(count=Freq));
run;

```

APPENDIX 3: USING PROC FREQ

```

/* Create a horizontal bar plot of tickets sold by different airlines */
proc sgplot data=airline_freq;
vbar airline /
response=Freq
datalabel
nostatlabel;
title 'Tickets Sold by Different Airlines';
xaxis label='Airlines';
yaxis label='Number of Tickets';
run;

/* Impact of buying tickets before departure */
/* Calculate average price and days_off by airline */
proc means data=project.flight noprint;
class airline;
var Price_US_dollars days_left;
output out=avg_airline(drop=_type_ _freq_) mean(Price_US_dollars)=avg_price mean(days_left)=avg_days_left;
run;

/* Check if prices varies for each price */
/* Create a bar plot of average price by airlines with different colors */
proc sgplot data=avg_airline;
vbar airline / response=avg_price
group=airline
datalabel
nostatlabel;
format avg_price dollar12.2;
title 'Average Price by Airlines in USD';
xaxis label='Airline';
yaxis label='Average Price';
run;

```

APPENDIX 4: USING PROC SGPLOT FOR VISUALIZATION

```

/* Plot bar plots for price vs. departure time and arrival time */
proc sgplot data=project.flight;
  vbar departure_time / response=Price_US_dollars stat=mean datalabel;
  xaxis label='Departure Time' fitpolicy=rotate;
  yaxis label='Price in USD';
  format Price_US_dollars dollar12.;
  title 'Average Price in USD by Departure Time';
run;

proc sgplot data=project.flight;
  vbar arrival_time / response=Price_US_dollars stat=mean datalabel;
  xaxis label='Departure Time' fitpolicy=rotate;
  yaxis label='Price in USD';
  format Price_US_dollars dollar12.;
  title 'Average Price in USD by Arrival Time';
run;

/* Plot KDE plots for price distribution by departure time */
proc sgpanel data=project.flight;
  panelby departure_time / layout=rowlattice columns=1 novarname;
  density Price_US_dollars / group=arrival_time;
  colaxis label='Price';
  rowaxis label='Density';
  format Price_US_dollars dollar12.;
  title 'Price Distribution by Departure Time';
run;

```

APPENDIX 5: USING PROC SGPANEL TO PLOT KDE PLOT.

```

/* Plot KDE plots for price distribution by arrival time */
proc sgpanel data=project.flight;
  panelby arrival_time / layout=rowlattice columns=1 novarname;
  density Price_US_dollars / group=departure_time;
  colaxis label='Price';
  rowaxis label='Density';
  format Price_US_dollars dollar12.;
  title 'Price Distribution by Arrival Time';
run;

/* Create donut chart for share of class */
ods graphics / reset width=7in height=4.5in;
title 'Flight Class Distribution';
proc sgpie data=project.flight;
  styleattrs datacolors=(gold olive);
  donut class / holevalue holelabel='Count' ringsize=0.5;
run;

/* Create KDE plot for price distribution by class */
proc sgpanel data=project.flight;
  panelby class / novarname columns=1;
  density Price_US_dollars / group=class;
  colaxis label='Price';
  rowaxis label='Density';
  format Price_US_dollars dollar12.;
  title 'Price Distribution by Class';
run;

```

APPENDIX 6: USING PROC SGPIE TO CREATE PIECHARTS


```

/* Create bar plot for number of stops vs. price */
proc sgplot data=project.flight;
  vbar stops / response=Price_US_dollars stat=mean group=class datalabel;
  xaxis label='Number of Stops';
  yaxis label='Price';
  format Price_US_dollars dollar12.;
  title 'Average Price by Number of Stops for both Business and Economy Class';
run;

/* Predicting Prices of Flight using Linear Regression Machine learning Model */
/* encode the Values */
data predictors;
  set project.flight;
  if class = 'Economy' then class_num = 0;
  else if class = 'Business' then class_num = 1;
  if airline = 'Indigo' then airline_num = 1;
  else if airline = 'Air_India' then airline_num = 2;
  else if airline = 'GO_FIRST' then airline_num = 3;
  else if airline = 'SpiceJet' then airline_num = 4;
  else if airline = 'Vistara' then airline_num = 5;
  else if airline = 'AirAsia' then airline_num = 6;
  if stops = 'zero' then stops_num = 0;
  else if stops = 'one' then stops_num = 1;
  else if stops = 'two' then stops_num = 2;
run;

/* Fit linear regression model with the right predictors */
proc reg data=predictors;
  model Price_US_dollars = airline_num class_num stops_num duration_minutes;
  title 'Linear Regression Model to Predict Price';
  output out=prediction_predicted predicted=predicted;
run;

```

APPENDIX 7: IMPLEMENTING PROC LINEAR REGRESSION

```

/* Printing predicted data */
proc print data=prediction_predicted (obs=20);
run;

/* Compute model performance metrics */
proc reg data=prediction_predicted;
  model Price_US_dollars = predicted;
  output out=prediction_metrics p=predicted_resid r=r;
run;

/* Print model performance metrics */
proc print data=prediction_metrics(obs=20);
  var Price_US_dollars predicted predicted_resid r;
run;

```

APPENDIX 8: EVALUATION OF THE PROC REG MODEL

jupyter Project Flight Pirce Prediction (1) Last Checkpoint: 2 minutes ago (autosaved)

File Edit View Insert Cell Kernel Widgets Help Not Trusted Python 3

In [13]:

```
import numpy as np #
import pandas as pd #
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score
import warnings
warnings.filterwarnings("ignore")
```

In [14]:

```
data = pd.read_csv('Clean_Dataset.csv', usecols=lambda col: col != 'Unnamed: 0')
```

In [15]:

```
data.head(20)
```

Out[15]:

	airline	flight	source_city	departure_time	stops	arrival_time	destination_city	class	duration	days_left	price
0	SpiceJet	SG-8709	Delhi	Evening	zero	Night	Mumbai	Economy	2.17	1	5953
1	SpiceJet	SG-8157	Delhi	Early_Morning	zero	Morning	Mumbai	Economy	2.33	1	5953
2	AirAsia	I5-764	Delhi	Early_Morning	zero	Early_Morning	Mumbai	Economy	2.17	1	5956
3	Vistara	UK-995	Delhi	Morning	zero	Afternoon	Mumbai	Economy	2.25	1	5955
4	Vistara	UK-963	Delhi	Morning	zero	Morning	Mumbai	Economy	2.33	1	5955
5	Vistara	UK-945	Delhi	Morning	zero	Afternoon	Mumbai	Economy	2.33	1	5955
6	Vistara	UK-927	Delhi	Morning	zero	Morning	Mumbai	Economy	2.08	1	6060
7	Vistara	UK-951	Delhi	Afternoon	zero	Evening	Mumbai	Economy	2.17	1	6060
8	GO_FIRST	G8-334	Delhi	Early_Morning	zero	Morning	Mumbai	Economy	2.17	1	5954

APPENDIX 9: IMPORTING THE DATASET IN PYTHON

In [16]:

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 300153 entries, 0 to 300152
Data columns (total 11 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   airline                               300153 non-null object
1   flight                                300153 non-null object
2   source_city                           300153 non-null object
3   departure_time                        300153 non-null object
4   stops                                 300153 non-null object
5   arrival_time                          300153 non-null object
6   destination_city                      300153 non-null object
7   class                                 300153 non-null object
8   duration                              300153 non-null float64
9   days_left                            300153 non-null int64
10  price                                 300153 non-null int64
dtypes: float64(1), int64(2), object(8)
memory usage: 25.2+ MB
```

APPENDIX 10: USING INFO() FUNCTION IN PYTHON

```
In [17]: data.describe()
```

```
Out[17]:
```

	duration	days_left	price
count	300153.000000	300153.000000	300153.000000
mean	12.221021	26.004751	20889.660523
std	7.191997	13.561004	22697.767366
min	0.830000	1.000000	1105.000000
25%	6.830000	15.000000	4783.000000
50%	11.250000	26.000000	7425.000000
75%	16.170000	38.000000	42521.000000
max	49.830000	49.000000	123071.000000

```
In [18]: data.isnull().sum()
```

```
Out[18]: airline          0  
flight                  0  
source_city             0  
departure_time          0  
stops                   0  
arrival_time            0  
destination_city        0  
class                   0  
duration                0  
days_left              0  
price                   0  
dtype: int64
```

APPENDIX 11: PERFORMING DATA CLEANING USING PYTHON

```
In [19]: #Convert the duration in hours to durations in minutes
data['duration_minutes'] = data['duration']*60
data.head(20)
```

Out[19]:

	airline	flight	source_city	departure_time	stops	arrival_time	destination_city	class	duration	days_left	price	duration_minutes
0	SpiceJet	SG-8709	Delhi	Evening	zero	Night	Mumbai	Economy	2.17	1	5953	130.2
1	SpiceJet	SG-8157	Delhi	Early_Morning	zero	Morning	Mumbai	Economy	2.33	1	5953	139.8
2	AirAsia	I5-764	Delhi	Early_Morning	zero	Early_Morning	Mumbai	Economy	2.17	1	5956	130.2
3	Vistara	UK-995	Delhi	Morning	zero	Afternoon	Mumbai	Economy	2.25	1	5955	135.0
4	Vistara	UK-963	Delhi	Morning	zero	Morning	Mumbai	Economy	2.33	1	5955	139.8
5	Vistara	UK-945	Delhi	Morning	zero	Afternoon	Mumbai	Economy	2.33	1	5955	139.8
6	Vistara	UK-927	Delhi	Morning	zero	Morning	Mumbai	Economy	2.08	1	6060	124.8
7	Vistara	UK-951	Delhi	Afternoon	zero	Evening	Mumbai	Economy	2.17	1	6060	130.2
8	GO_FIRST	G8-334	Delhi	Early_Morning	zero	Morning	Mumbai	Economy	2.17	1	5954	130.2
9	GO_FIRST	G8-336	Delhi	Afternoon	zero	Evening	Mumbai	Economy	2.25	1	5954	135.0
10	GO_FIRST	G8-392	Delhi	Afternoon	zero	Evening	Mumbai	Economy	2.25	1	5954	135.0
11	GO_FIRST	G8-338	Delhi	Morning	zero	Afternoon	Mumbai	Economy	2.33	1	5954	139.8
12	Indigo	6E-5001	Delhi	Early_Morning	zero	Morning	Mumbai	Economy	2.17	1	5955	130.2
13	Indigo	6E-6202	Delhi	Morning	zero	Afternoon	Mumbai	Economy	2.17	1	5955	130.2
14	Indigo	6E-549	Delhi	Afternoon	zero	Evening	Mumbai	Economy	2.25	1	5955	135.0
15	Indigo	6E-6278	Delhi	Morning	zero	Morning	Mumbai	Economy	2.33	1	5955	139.8
16	Air_India	AI-887	Delhi	Early_Morning	zero	Morning	Mumbai	Economy	2.08	1	5955	124.8
17	Air_India	AI-665	Delhi	Early_Morning	zero	Morning	Mumbai	Economy	2.17	1	5955	130.2

APPENDIX 12: PERFORMING DATA PREPROCESSING USING PYTHON

```
In [22]: # Q1 Does price vary with Airlines?
#create a barplot
plt.figure(figsize=(10, 6))
sns.barplot(x='airline', y='price_usd', data=data,errorbar=None, palette='muted')
plt.title('Average Price by Airlines')
plt.xlabel('Airline')
plt.ylabel('Average Price (USD)')
plt.xticks(rotation=45) # Rotate x-axis labels for better readability
plt.show()
```

APPENDIX 13: VISUALIZATION USING PYTHON

```

# Plot KDE plot for price distribution by source city
sns.kdeplot(x='price_usd', hue='source_city', data=data, ax=ax1)
ax1.set_title('Price Distribution by Source City', fontsize=16)
ax1.set_xlabel('Price (USD)', fontsize=16)
ax1.set_ylabel('Density', fontsize=16)

# Plot KDE plot for price distribution by destination city
sns.kdeplot(x='price_usd', hue='destination_city', data=data, ax=ax2)
ax2.set_title('Price Distribution by Destination City', fontsize=16)
ax2.set_xlabel('Price (USD)', fontsize=16)
ax2.set_ylabel('Density', fontsize=16)

# Integrate insightful analysis into the code comments
"""
These KDE plots show the distribution of ticket prices based on the source and destination cities:

- The distributions indicate that ticket prices vary significantly based on both source and destination cities.
- The differences in price distribution could be attributed to factors such as distance, popularity of routes,
and demand-supply dynamics.
"""

plt.tight_layout()
plt.show()

```

APPENDIX 14: CREATING A KDE PLOT USING PYTHON

MACHINE LEARNING ALGORITHM

linear Regression Algorithm will be made use of in this project for predicting flight prices

```

In [30]: # Encode categorical variables
label_encoder = LabelEncoder()
data['airline_num'] = label_encoder.fit_transform(data['airline'])
data['class_num'] = label_encoder.fit_transform(data['class'])
data['stops_num'] = label_encoder.fit_transform(data['stops'])

In [31]: # Define predictors and target variable
X = data[['airline_num', 'class_num', 'stops_num', 'duration_minutes']]
y = data['price_usd']

In [32]: # Split the data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

In [33]: # Train the model
model = LinearRegression()
model.fit(X_train, y_train)

```

APPENDIX 15: IMPLEMENTATION OF LINEAR REGRESSION USING PYTHON

```

1 print(f"Train Mean Squared Error: {train_mse:.2f}")
2 print(f"Test Mean Squared Error: {test_mse:.2f}")
3 print(f"Train Mean Absolute Error: {train_mae:.2f}")
4 print(f"Test Mean Absolute Error: {test_mae:.2f}")
5 print(f"Train R-squared: {train_r2:.2f}")
6 print(f"Test R-squared: {test_r2:.2f}")

```

Train Mean Squared Error: 8940.32
 Test Mean Squared Error: 8905.93
 Train Mean Absolute Error: 58.85
 Test Mean Absolute Error: 58.51
 Train R-squared: 0.88
 Test R-squared: 0.88

APPENDIX 16: RESULTS OF LINEAR REGRESSION MODEL USING PYTHON

Training Data - Actual vs Predicted Price:

	Actual Price (USD)	Predicted Price (USD)	Price Difference (USD)
148417	162.288	78.907257	83.380743
36879	119.280	78.907257	40.372743
274531	671.796	630.481390	41.314610
166397	95.124	78.907257	16.216743
272722	666.024	630.481390	35.542610
183550	62.472	78.907257	-16.435257
47422	62.784	78.907257	-16.123257
255047	746.256	630.481390	115.774610
200793	87.744	78.907257	8.836743
37174	51.012	78.907257	-27.895257
190559	55.644	78.907257	-23.263257
43344	180.936	78.907257	102.028743
176994	81.036	78.907257	2.128743
2038	150.840	78.907257	71.932743
117329	79.032	78.907257	0.124743
168414	80.808	78.907257	1.900743
3054	69.672	78.907257	-9.235257
228452	493.212	630.481390	-137.269390
40589	47.952	78.907257	-30.955257
220730	566.604	630.481390	-63.877390

APPENDIX 17: COMPARING THE ACTUAL PRICE AGAINST PREDICTED PRICE

#

Logistics Regression Modelling

```
[28]: 1 # Define predictors and target variable
      2 X = data[['price_usd', 'class_num']]
      3 y = data['stops_num']

[29]: 1 # Split the data into training and testing sets
      2 X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

[30]: 1 model = LogisticRegression()
      2 model.fit(X_train, y_train)
      3

t[30]: ▾ LogisticRegression
      LogisticRegression()
```

APPENDIX 18: IMPLEMENTING LOGISTICS REGRESSION USING PYTHON

```
Training Metrics:
Accuracy: 0.8533911928103214
Precision: 0.6303869089783963
Recall: 0.39542718398751875
F1 Score: 0.4105743857512259
Confusion Matrix:
[[199391    53   1126]
 [ 10631    14     20]
 [ 23374     0   5513]]

Test Metrics:
Accuracy: 0.8545085039396312
Precision: 0.5730441084751411
Recall: 0.3929420417538653
F1 Score: 0.40718553750732084
Confusion Matrix:
[[49982    20    291]
 [ 2614     1     6]
 [ 5803     0   1314]]
```

APPENDIX 19: EVALUATION OF LOGISTICS REGRESSION MODEL USING PYTHON

Training Data - Actual vs Predicted Stops Number:

	Actual Stops Number	Predicted Stops Number	Mismatch
148417	0	0	False
36879	1	0	True
274531	0	0	False
166397	0	0	False
272722	0	0	False
183550	0	0	False
47422	0	0	False
255047	0	0	False
200793	0	0	False
37174	0	0	False
190559	0	0	False
43344	0	0	False
176994	0	0	False
2038	0	0	False
117329	0	0	False
168414	0	0	False
3054	0	0	False
228452	0	0	False
40589	0	0	False
220730	0	0	False

APPENDIX: 20 COMPARING ACTUALS STOPS AGAINST PREDICTED STOPS

Peer Evaluation Form for Group Work

Your Group ID: 56

Write the **Student ID** of each of your group members in a separate column. For each group member, indicate the extent to which you agree with the statement on the left, using a scale of 1-5:

1 = lowest score, 5 = highest score

- 1: Very poor: unacceptable performance
- 2: poor: less than acceptable performance
- 3: Average performance
- 4: Good performance
- 5: Excellent performance

Evaluation Criteria	Group member 1: ID	Group member 2: ID	Group member 3: ID	Group member 4: ID	Comment
	100701434	100671487	100724916	100685307	
	Grade: (1 - 5)	Grade: (1 - 5)	Grade: (1 - 5)	Grade: (1 - 5)	
•Did the individual contribute his/her fair share? •Contributes meaningfully to group discussions. •Contributes significantly to the success of the project. •Demonstrates a cooperative and supportive attitude. •Overall, how would you rank this person's contributions to the group?	5	5	5	5	Every member of the group contributed to the final output of the project.
•Did the individual complete all work in a timely manner?	5	5	5	5	Yes, we all did.
•How would you rate the quality of individuals' work?	5	5	5	5	Excellent

•Did the individual maintain a positive, respectful attitude? •Attends group meetings regularly and arrives on time. •Would you want to work with this person again?	5	5	5	5	Yes, we would love to work with each other again.
TOTALS	20	20	20	20	
PEMark = (TOTALS / 20)	1	1	1	1	
PEMark: peer evaluation mark ** If a student <u>never</u> joined the group, put “ <i>no participation</i> ” in the corresponding column.					

Overall mark for a student (CW) = CW mark * peer evaluation mark